1.3 Microprocessors at work

In this lesson

You will learn:

- how microprocessors are used to improve the way people work.



Doctors

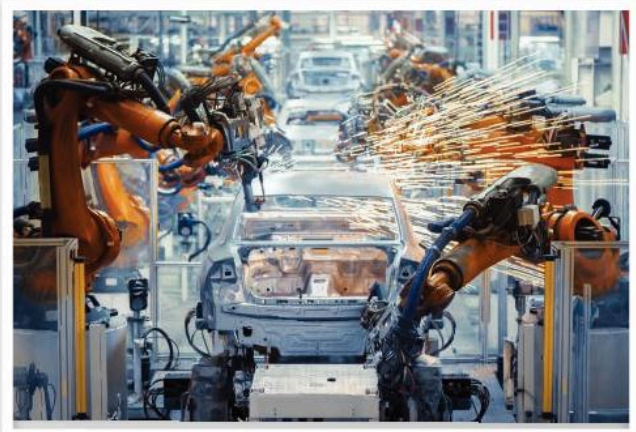
Doctors diagnose illness using machines that contain microprocessors. Two important machines are magnetic resonance imaging (MRI) and computerised axial tomography (CAT) scanners. MRI and CAT scanners create a 3D picture of the inside of a patient. A doctor can use the picture to see problems.

Technology also helps patients to recover after treatment. Patient monitoring systems check blood pressure, temperature and pulse rate. Nurses get an early warning if a patient needs attention.

Being a doctor is a complex job. There is a lot to know about many illnesses. Doctors use the internet for research. They also use the internet to contact experts. Sometimes those experts work in another country.

Manufacturing

A modern car factory is full of technology. Robots are used to assemble cars. A car is moved automatically from one part of the factory to another. At each stop, robots complete work on part of the car. Robots do work that used to be done by humans.



Robots are used to manufacture many other goods. They are used for simple repetitive jobs. Robots are also used to do jobs that are dangerous for humans. For example, the police use robots to investigate packages that might contain explosives.

Retail



Supermarkets and other large shops depend on technology. At the supermarket checkout, goods are scanned using a barcode reader. This creates a detailed receipt for the customer. Information gathered at the checkout is used by the supermarket to decide what items they need to order from their suppliers. Customers pay for their shopping using debit and credit cards. Money is transferred automatically from the customer to the shop.

Many people now use the internet to do their shopping. Orders made on the internet are processed quickly by computer systems. Goods are often delivered on the day after an order is made. Some people worry about the effect internet shopping has on shops in town centres.



Activity

Working in a group, interview your teacher to find out how using technology improves their job. Work together to write a list of three questions before you start the interview.



Extra challenge

Jobs that used to be done by humans are being replaced by automation and robots. Talk about this with classmates and your family. What are the good and bad things about automation?

Think again

Do you think your teacher could be replaced by a robot? Give reasons for your answer. How would you feel about travelling to school in a bus driven by a robot?

1.4 Computer storage

In this lesson

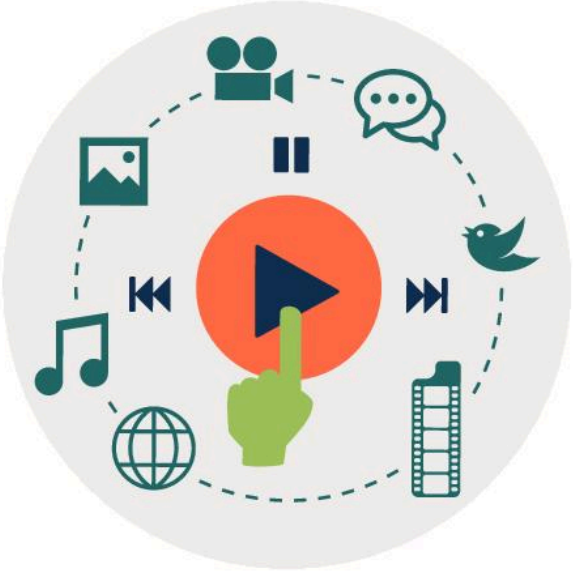
You will learn:

- about computer storage
- that computer storage is an important component of a computer system
- why computer storage is important.

Data files

Different types of media can be created and used on a computer. The media types used on computers include:

- text documents (for example, homework or assignments)
- pictures and photographs
- videos
- music



All types of media used on a computer are stored in files. A file stored on a computer is called a **data file**.

Computer storage

When you create work on a computer, it is important to save it correctly. Work saved in a data file can be used over and over again. If your work is not saved correctly it can be lost.

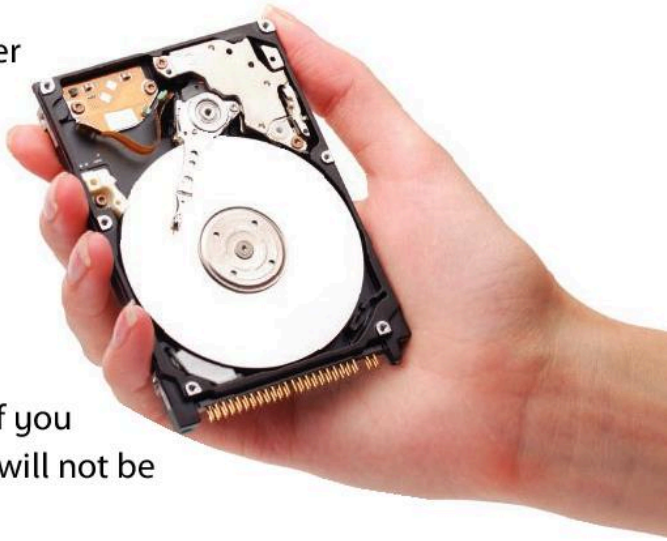
All computers have a place to store data files. It is called computer storage.

Where data files are stored

A computer uses a device called a **storage drive** to save files. Every computer has a storage drive built into the case. A storage drive can save a large amount of information.

For example, a storage drive used in a desktop computer can store:

- 150,000 photographs or
- 8500 hours of music or
- 300,000 books.



Network storage

You will probably not always use the same computer. If you save work onto the storage drive of one computer, you will not be able to use that file if you move to another computer.

Computers in schools and offices are usually connected. We call a group of connected computers a **network**. A network has its own storage drive. Everyone using a network has a special area on the **network storage** drive where they can save their work. If you save your work on your school network drive, you can use it on any computer that is connected to your school network.

Flash drives

A **flash drive** is a small portable storage device. It is small enough to fit in your pocket and can be used to move files from one computer to another. A flash drive is plugged into a computer so that a data file can be saved to it.





Activity

How could you use a flash drive to move homework from a computer at home to a computer at school? Write your answer as a list of steps.



Extra challenge

You have learned that a typical storage drive can hold 8500 hours of music. How many days would it take to play that music? (Hint: there are 24 hours in a day.)

Think again

What problems will you have if you don't save your work properly?